Sohil Khan

Email : khansohil530@gmail.com
Github Website LinkedIn
Mobile : +91-9887581808

Backend-focused Software Engineer with 2+ years of experience designing distributed systems, data migration pipelines, and secure integrations, delivering high-impact solutions at scale in cloud-native environments.

Experience

Alphasense — New Delhi, IND
Software Engineer 2 — *Jun 2022 – Nov 2024*

- **Migration System**:
  - Engineered a scalable migration framework on **AWS EC2** with **Python** and **MongoDB**, enabling Sales team to independently manage client migrations through a Retool UI without developer intervention.
  - Automated the migration of **100M+ rows**, reducing the timeline by 6 months by streamlining customer feedback loops to function without developer involvement.
  - Reduced unnecessary transfers by **30%**, accelerated customer adoption, and ensured uninterrupted operations while migrating **1,300+ customer** accounts, driving **$400K ARR** growth.
- **Integrations**:
  - Worked on development of connectors to enable synchronization of client content form third-party vendors like Box.
  - Optimized Python workers to scale ingestion rate by introducing batching and parallel processing.
  - Architected a microservices-based system with **FastAPI** APIs (**Uvicorn**) and Python workers (**AWS SQS**, **MySQL** metadata), delivering reliable synchronization with less than 1% failure rate.
  - Implemented observability with **OpenTelemetry** + **GrafanaTempo**, improving monitoring, troubleshooting, and uptime consistency.
- **Jupyterhub**:
  - Deployed a secure JupyterHub environment on Docker Swarm with **RBAC** and Retool-based user provisioning, providing **30** developers across 3 teams controlled backend access.
  - Reduced customer query turnaround from **3–4 days to 2–5 hours**, significantly improving customer satisfaction and reducing support backlog.
  - Increased developer efficiency by **70%** and boosted customer support productivity by **80%** by enabling shared, reusable notebooks for recurring requests.
- **Secure Token Migration**:
  - Migrated all third-party client tokens to an **AES-256** encrypted framework with **Vault** integration, reducing exposure risk by **90%**, achieving **ISO 27001** compliance, and delivering the project end-to-end as a solo contributor.
  - Refactored integration code across connectors to support encrypted tokens with zero downtime, ensuring business continuity for all clients.
  - Directly contributed to onboarding major enterprise clients by addressing strict security requirements, earning recognition from leadership for securing high-value deals.

Programming Skills

- **Languages**: Python, SQL, JavaScript
- **Tools**: MySQL, MongoDB, Redis, AWS, Docker, Kubernetes, Docker Swarm, Vault, OpenTelemetry, GrafanaTempo
- **Skills**: Distributed Systems, Microservices Architecture, API Development, Data Synchronization
- **Course Work**: Database Management System, Operating Systems, Computer Networking

Education

- **Birla Institute of Technology and Science** — Pilani, India
  B.E Engineering in Electrical and Electronics — *Aug 2017 – July 2022*
- **Birla Institute of Technology and Science** — Pilani, India
  M.Sc Economic — *Aug 2017 – July 2022*